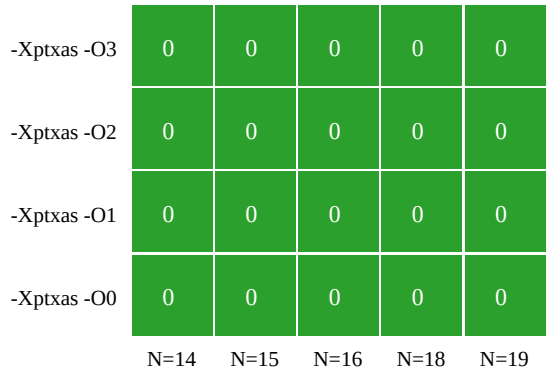
clmad audit on sm_120: miscompile reproduction, ecc2k130 tests, two-class timing

research/clmad_audit_20261110 · protocol frozen before data · every number from results/summary.json · claims [pcc:unverified] until the benchmark record is closed and validated

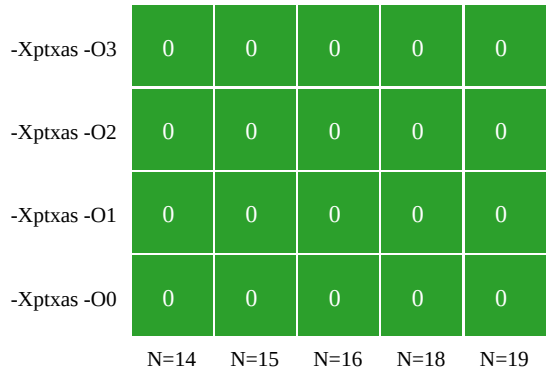
arm	GPU	driver	nvcc	pod	run exit	Q1	Q2 host / device	Q3 latency / occupancy
cuda13.3.1	NVIDIA RTX PRO 6000 Blackwell Server Edition cc 12.0	595.91.07 CUDA 13.3	13.3.73	84o1aurclqz4mj	0	not reproduced	2 / 0	no class distinguishable no class distinguishable
cuda13.4.3	NVIDIA GeForce RTX 5090 cc 12.0	595.91.07 CUDA 13.4	13.4.92	hm5x2mcbavf75w	0	not reproduced	2 / 0	no class distinguishable no class distinguishable

cuda13.3.1: repro.cu exit by N and ptxas level (0 match, 1 mismatch, 2 CUDA error)



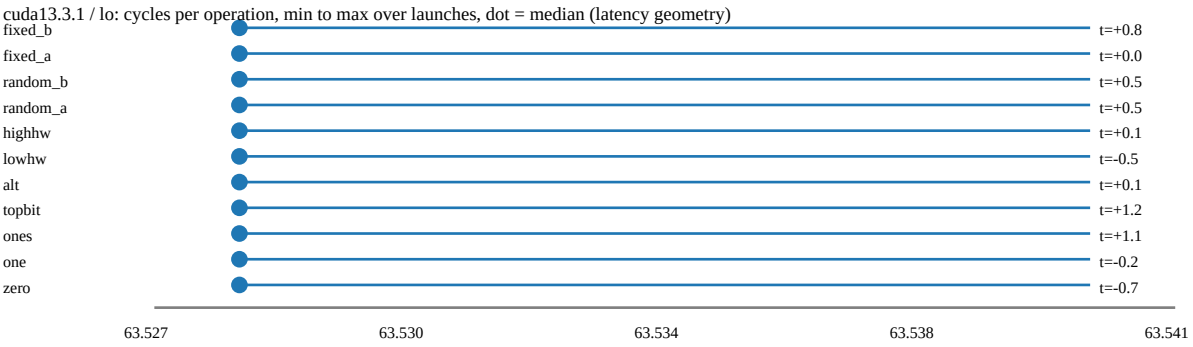
cuda13.3.1: Q1 not reproduced []; controls {'karatsuba_ptxjit': '0', 'karatsuba_seed2': '0', 'schoolbook': '0', 'square': '0'}. SASS CLMAD counts per binary: repro_N14_O3=78, repro_N14_O2=78, repro_N14_O1=78, repro_N14_O0=84, repro_N15_O3=84, repro_N15_O2=84, repro_N15_O1=84, repro_N15_O0=90, repro_N16_O3=90, repro_N16_O2=90, repro_N16_O1=90, repro_N16_O0=96, repro_N18_O3=102, repro_N18_O2=102, repro_N18_O1=102, repro_N18_O0=108, repro_N19_O3=108, repro_N19_O2=108, repro_N19_O1=108, repro_N19_O0=114, repro_N14_O3_schoolbook=104, repro_N64_O3_square=384, repro_N14_O3_seed2=78

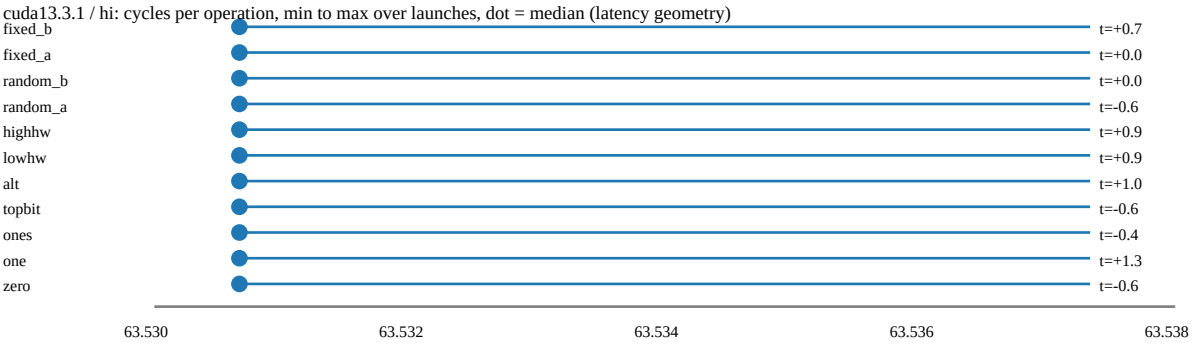
cuda13.4.3: repro.cu exit by N and ptxas level (0 match, 1 mismatch, 2 CUDA error)

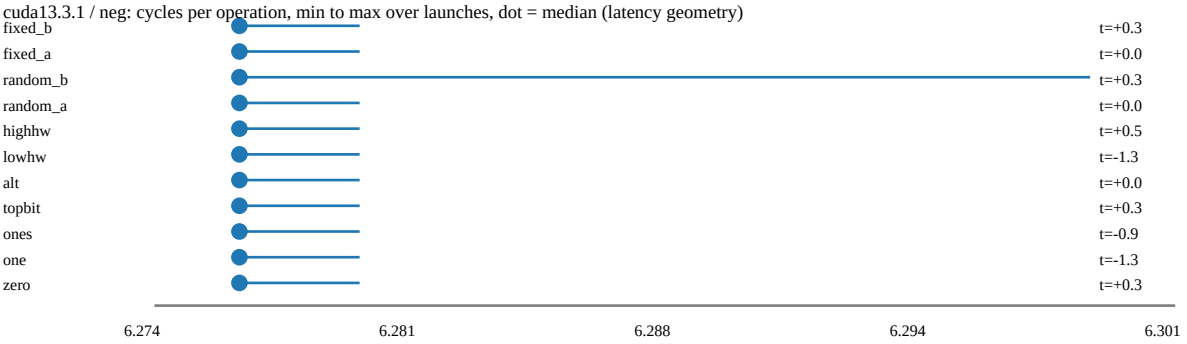


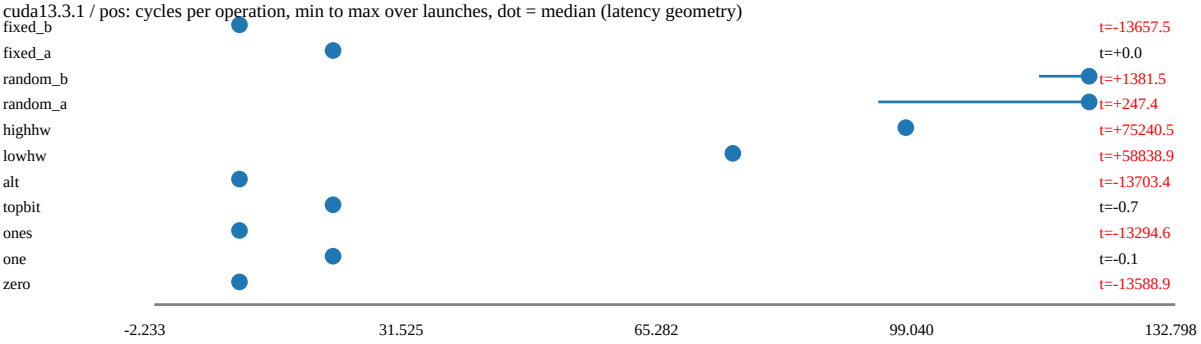
cuda13.4.3: Q1 not reproduced []; controls {'karatsuba_ptxjit': '0', 'karatsuba_seed2': '0', 'schoolbook': '0', 'square': '0'}. SASS CLMAD counts per binary: repro_N14_O3=78, repro_N14_O2=78, repro_N14_O1=78, repro_N14_O0=84, repro_N15_O3=84, repro_N15_O2=84, repro_N15_O1=84, repro_N15_O0=90, repro_N16_O3=90, repro_N16_O2=90, repro_N16_O1=90, repro_N16_O0=96, repro_N18_O3=102, repro_N18_O2=102, repro_N18_O1=102, repro_N18_O0=108, repro_N19_O3=108, repro_N19_O2=108, repro_N19_O1=108, repro_N19_O0=114, repro_N14_O3_schoolbook=104, repro_N64_O3_square=384, repro_N14_O3_seed2=78

cuda13.3.1: timing. Positive control t = -23189; negative control max |t| = 1.30; threshold 4.5.









cuda13.3.1 / latency geometry ({'blocks': 1, 'cc': '12.0', 'chain': 2048, 'device': 'NVIDIA RTX PRO 6000 Blackwell Server Edition', 'geometry': 'latency', 'reps': 100, 'seed': 20261110, 'sms': 188, 'threads': 32})

mode	class	n	median c/op	min	max	sd	t vs fixed_a
lo	zero	100	63.528	63.528	63.540	0.0038	-0.67
lo	one	100	63.528	63.528	63.540	0.0039	-0.18
lo	ones	100	63.528	63.528	63.540	0.0041	+1.14
lo	topbit	100	63.528	63.528	63.540	0.0040	+1.17
lo	alt	100	63.528	63.528	63.540	0.0038	+0.11
lo	lowhw	100	63.528	63.528	63.540	0.0039	-0.47
lo	highhw	100	63.528	63.528	63.540	0.0040	+0.06

mode	class	n	median c/op	min	max	sd	t vs fixed_a
lo	random_a	100	63.528	63.528	63.540	0.0041	+0.49
lo	random_b	100	63.528	63.528	63.540	0.0040	+0.51
lo	fixed_a	100	63.528	63.528	63.540	0.0039	+0.00
lo	fixed_b	100	63.528	63.528	63.540	0.0043	+0.76
lo	A/A random_a~random_b		t -0.02		fixed pair	t -0.76	low~high t -0.5
hi	zero	100	63.530	63.530	63.537	0.0025	-0.56
hi	one	100	63.530	63.530	63.537	0.0030	+1.34
hi	ones	100	63.530	63.530	63.537	0.0026	-0.37
hi	topbit	100	63.530	63.530	63.537	0.0025	-0.56
hi	alt	100	63.530	63.530	63.537	0.0030	+1.02
hi	lowhw	100	63.530	63.530	63.537	0.0029	+0.86
hi	highhw	100	63.530	63.530	63.537	0.0029	+0.86
hi	random_a	100	63.530	63.530	63.537	0.0025	-0.56
hi	random_b	100	63.530	63.530	63.537	0.0027	+0.00
hi	fixed_a	100	63.530	63.530	63.537	0.0027	+0.00
hi	fixed_b	100	63.530	63.530	63.537	0.0029	+0.69
hi	A/A random_a~random_b		t -0.56		fixed pair	t -0.69	low~high t +0.0
neg	zero	100	6.276	6.276	6.280	0.0009	+0.27
neg	one	100	6.276	6.276	6.280	0.0005	-1.30
neg	ones	100	6.276	6.276	6.280	0.0006	-0.93
neg	topbit	100	6.276	6.276	6.280	0.0009	+0.27
neg	alt	100	6.276	6.276	6.280	0.0008	+0.00
neg	lowhw	100	6.276	6.276	6.280	0.0005	-1.30
neg	highhw	100	6.276	6.276	6.280	0.0009	+0.52
neg	random_a	100	6.276	6.276	6.280	0.0008	+0.00
neg	random_b	100	6.276	6.276	6.299	0.0023	+0.27
neg	fixed_a	100	6.276	6.276	6.280	0.0008	+0.00
neg	fixed_b	100	6.276	6.276	6.280	0.0009	+0.27
neg	A/A random_a~random_b		t -0.27		fixed pair	t -0.27	low~high t -1.8
pos	zero	100	9.026	9.020	9.040	0.0063	-13588.86
pos	one	100	21.401	21.395	21.415	0.0061	-0.08
pos	ones	100	9.026	9.020	9.041	0.0066	-13294.61
pos	topbit	100	21.401	21.395	21.415	0.0058	-0.70
pos	alt	100	9.027	9.020	9.041	0.0062	-13703.42
pos	lowhw	100	74.283	74.262	74.291	0.0062	+58838.88
pos	highhw	100	97.164	97.145	97.172	0.0076	+75240.49
pos	random_a	100	121.414	93.520	121.545	3.9945	+247.40
pos	random_b	100	121.414	114.789	121.422	0.7195	+1381.48
pos	fixed_a	100	21.405	21.395	21.416	0.0065	+0.00
pos	fixed_b	100	9.027	9.020	9.040	0.0063	-13657.48

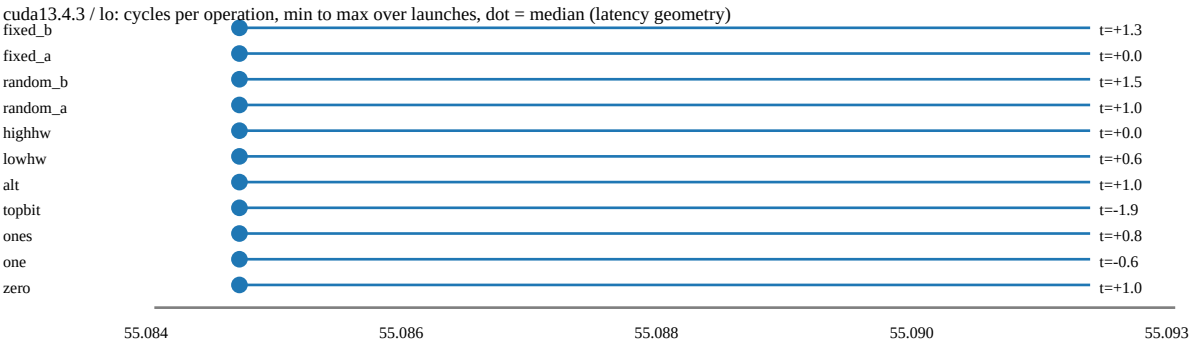
mode	class	n	median c/op	min	max	sd	t vs fixed_a
pos	A/A random_a~random_b		t -1.42		fixed pair	t +13657.48	low~high t -23189.2

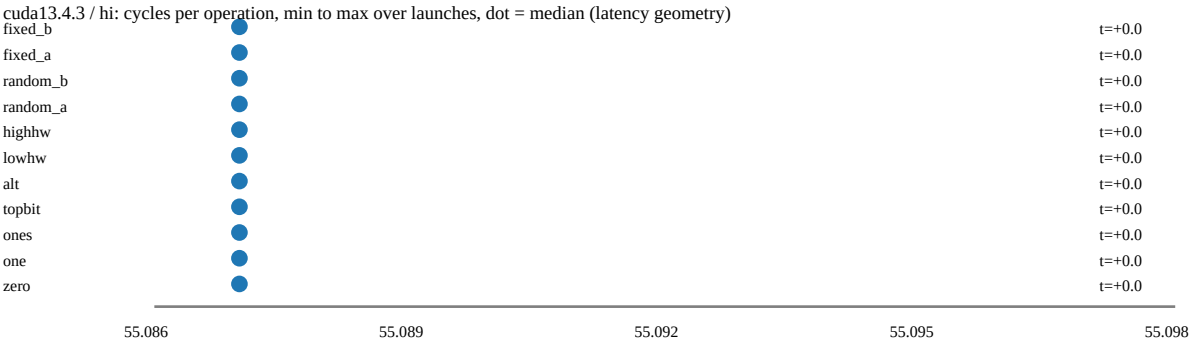
cuda13.3.1 / occupancy geometry ({'blocks': 376, 'cc': '12.0', 'chain': 2048, 'device': 'NVIDIA RTX PRO 6000 Blackwell Server Edition', 'geometry': 'occupancy', 'reps': 50, 'seed': 20261111, 'sms': 188, 'threads': 256})

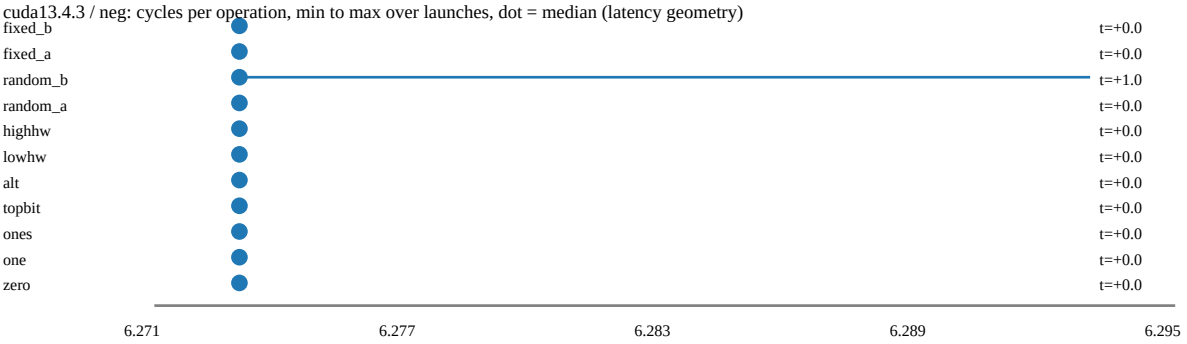
mode	class	n	median c/op	min	max	sd	t vs fixed_a
lo	zero	50	255.901	255.896	255.929	0.0088	-0.18
lo	one	50	255.909	255.896	255.923	0.0072	+1.44
lo	ones	50	255.902	255.894	255.928	0.0093	+0.40
lo	topbit	50	255.908	255.896	255.928	0.0089	+1.38
lo	alt	50	255.902	255.897	255.933	0.0096	+0.91
lo	lowhw	50	255.901	255.895	255.924	0.0074	-0.51
lo	highhw	50	255.905	255.896	255.918	0.0060	-0.49
lo	random_a	50	255.899	255.894	255.930	0.0075	-1.55
lo	random_b	50	255.904	255.892	255.920	0.0076	-0.09
lo	fixed_a	50	255.901	255.896	255.928	0.0081	+0.00
lo	fixed_b	50	255.907	255.896	255.928	0.0084	+1.32
lo	A/A random_a~random_b		t -1.50		fixed pair	t -1.32	low~high t -0.1
hi	zero	50	255.905	255.899	255.931	0.0085	+0.27
hi	one	50	255.907	255.899	255.932	0.0073	+0.40
hi	ones	50	255.904	255.897	255.930	0.0093	+0.23
hi	topbit	50	255.904	255.896	255.933	0.0098	+0.59
hi	alt	50	255.903	255.899	255.931	0.0072	-0.97
hi	lowhw	50	255.905	255.898	255.931	0.0080	+0.10
hi	highhw	50	255.905	255.899	255.932	0.0075	-0.14
hi	random_a	50	255.901	255.896	255.930	0.0082	-1.48
hi	random_b	50	255.903	255.895	255.933	0.0080	-1.01
hi	fixed_a	50	255.904	255.898	255.932	0.0086	+0.00
hi	fixed_b	50	255.905	255.897	255.930	0.0075	-0.83
hi	A/A random_a~random_b		t -0.50		fixed pair	t +0.83	low~high t +0.2
neg	zero	50	13.216	13.196	13.240	0.0116	+2.91
neg	one	50	13.209	13.196	13.241	0.0095	+1.21
neg	ones	50	13.207	13.184	13.242	0.0127	+0.70
neg	topbit	50	13.211	13.196	13.241	0.0124	+1.83
neg	alt	50	13.208	13.195	13.240	0.0104	+0.17
neg	lowhw	50	13.210	13.199	13.241	0.0104	+1.54
neg	highhw	50	13.214	13.197	13.241	0.0108	+2.60
neg	random_a	50	13.206	13.195	13.237	0.0103	-0.27
neg	random_b	50	13.209	13.196	13.237	0.0106	+1.38
neg	fixed_a	50	13.206	13.186	13.245	0.0119	+0.00
neg	fixed_b	50	13.206	13.196	13.235	0.0097	+0.07

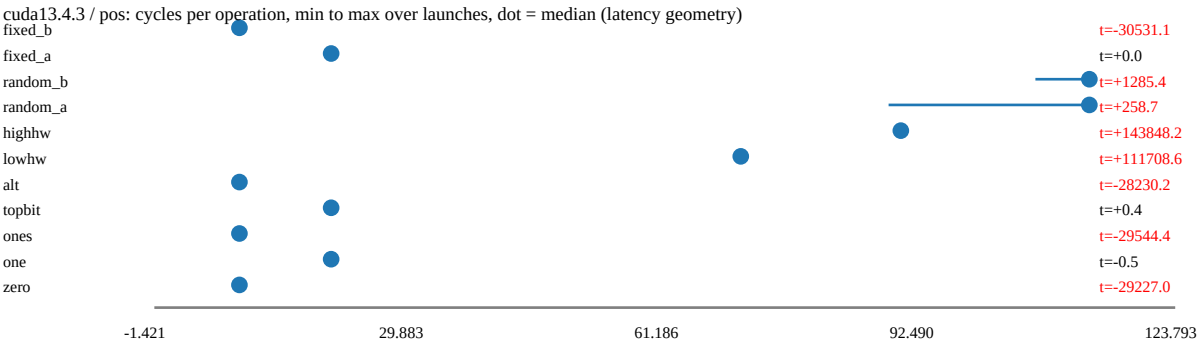
mode	class	n	median c/op	min	max	sd	t vs fixed_a
neg	A/A random_a~random_b		t -1.78		fixed pair	t -0.07	low~high t -1.2
pos	zero	50	31.962	31.946	31.985	0.0106	+0.89
pos	one	50	104.954	104.858	105.527	0.2492	+2052.73
pos	ones	50	31.965	31.943	31.979	0.0099	+1.36
pos	topbit	50	104.949	104.830	105.490	0.2510	+2038.98
pos	alt	50	31.955	31.945	31.983	0.0093	-0.90
pos	lowhw	50	257.377	257.336	257.413	0.0181	+76617.97
pos	highhw	50	353.491	353.372	353.659	0.0682	+32648.10
pos	random_a	50	450.944	450.691	451.115	0.0964	+30267.90
pos	random_b	50	450.953	450.731	451.141	0.0873	+33405.00
pos	fixed_a	50	31.959	31.945	31.982	0.0098	+0.00
pos	fixed_b	50	31.955	31.944	31.981	0.0107	-0.19
pos	A/A random_a~random_b		t -0.65		fixed pair	t +0.19	low~high t -9528.1

cuda13.4.3: timing. Positive control t = -36705; negative control max |t| = 1.00; threshold 4.5.









cuda13.4.3 / latency geometry ({'blocks': 1, 'cc': '12.0', 'chain': 2048, 'device': 'NVIDIA GeForce RTX 5090', 'geometry': 'latency', 'reps': 100, 'seed': 20261110, 'sms': 170, 'threads': 32})

mode	class	n	median c/op	min	max	sd	t vs fixed_a
lo	zero	100	55.084	55.084	55.092	0.0022	+1.04
lo	one	100	55.084	55.084	55.092	0.0014	-0.65
lo	ones	100	55.084	55.084	55.092	0.0021	+0.80
lo	topbit	100	55.084	55.084	55.092	0.0007	-1.93
lo	alt	100	55.084	55.084	55.092	0.0022	+1.04
lo	lowhw	100	55.084	55.084	55.092	0.0020	+0.55
lo	highhw	100	55.084	55.084	55.092	0.0017	+0.00

mode	class	n	median c/op	min	max	sd	t vs fixed_a
lo	random_a	100	55.084	55.084	55.092	0.0022	+1.04
lo	random_b	100	55.084	55.084	55.092	0.0024	+1.48
lo	fixed_a	100	55.084	55.084	55.092	0.0017	+0.00
lo	fixed_b	100	55.084	55.084	55.092	0.0023	+1.27
lo	A/A random_a~random_b		t -0.45		fixed pair	t -1.27	low~high t +0.6
hi	zero	100	55.087	55.087	55.087	0.0000	+0.00
hi	one	100	55.087	55.087	55.087	0.0000	+0.00
hi	ones	100	55.087	55.087	55.087	0.0000	+0.00
hi	topbit	100	55.087	55.087	55.087	0.0000	+0.00
hi	alt	100	55.087	55.087	55.087	0.0000	+0.00
hi	lowhw	100	55.087	55.087	55.087	0.0000	+0.00
hi	highhw	100	55.087	55.087	55.087	0.0000	+0.00
hi	random_a	100	55.087	55.087	55.087	0.0000	+0.00
hi	random_b	100	55.087	55.087	55.087	0.0000	+0.00
hi	fixed_a	100	55.087	55.087	55.087	0.0000	+0.00
hi	fixed_b	100	55.087	55.087	55.087	0.0000	+0.00
hi	A/A random_a~random_b		t +0.00		fixed pair	t +0.00	low~high t +0.0
neg	zero	100	6.273	6.273	6.273	0.0000	+0.00
neg	one	100	6.273	6.273	6.273	0.0000	+0.00
neg	ones	100	6.273	6.273	6.273	0.0000	+0.00
neg	topbit	100	6.273	6.273	6.273	0.0000	+0.00
neg	alt	100	6.273	6.273	6.273	0.0000	+0.00
neg	lowhw	100	6.273	6.273	6.273	0.0000	+0.00
neg	highhw	100	6.273	6.273	6.273	0.0000	+0.00
neg	random_a	100	6.273	6.273	6.273	0.0000	+0.00
neg	random_b	100	6.273	6.273	6.293	0.0019	+1.00
neg	fixed_a	100	6.273	6.273	6.273	0.0000	+0.00
neg	fixed_b	100	6.273	6.273	6.273	0.0000	+0.00
neg	A/A random_a~random_b		t -1.00		fixed pair	t +0.00	low~high t +0.0
pos	zero	100	9.014	9.014	9.021	0.0027	-29227.01
pos	one	100	20.264	20.264	20.271	0.0026	-0.54
pos	ones	100	9.014	9.014	9.021	0.0026	-29544.42
pos	topbit	100	20.264	20.264	20.271	0.0029	+0.38
pos	alt	100	9.014	9.014	9.021	0.0029	-28230.18
pos	lowhw	100	70.506	70.506	70.521	0.0035	+111708.57
pos	highhw	100	90.139	90.139	90.153	0.0040	+143848.20
pos	random_a	100	113.264	88.639	113.359	3.5523	+258.74
pos	random_b	100	113.264	106.646	113.271	0.7190	+1285.44
pos	fixed_a	100	20.264	20.264	20.271	0.0027	+0.00
pos	fixed_b	100	9.014	9.014	9.021	0.0024	-30531.12

mode	class	n	median c/op	min	max	sd	t vs fixed_a
pos	A/A random_a~random_b		t -1.41		fixed pair	t +30531.12	low~high t -36704.7

cuda13.4.3 / occupancy geometry ({'blocks': 340, 'cc': '12.0', 'chain': 2048, 'device': 'NVIDIA GeForce RTX 5090', 'geometry': 'occupancy', 'reps': 50, 'seed': 20261111, 'sms': 170, 'threads': 256})

mode	class	n	median c/op	min	max	sd	t vs fixed_a
lo	zero	50	255.856	255.840	255.894	0.0089	-0.88
lo	one	50	255.861	255.844	255.916	0.0145	+1.50
lo	ones	50	255.860	255.846	255.912	0.0132	+1.12
lo	topbit	50	255.860	255.842	255.919	0.0180	+1.65
lo	alt	50	255.860	255.845	255.914	0.0156	+1.81
lo	lowhw	50	255.861	255.844	255.923	0.0209	+2.04
lo	highhw	50	255.859	255.844	255.913	0.0142	+1.03
lo	random_a	50	255.857	255.841	255.921	0.0187	+1.12
lo	random_b	50	255.856	255.841	255.919	0.0160	-0.02
lo	fixed_a	50	255.860	255.842	255.881	0.0097	+0.00
lo	fixed_b	50	255.859	255.844	255.902	0.0131	+1.56
lo	A/A random_a~random_b		t +0.97		fixed pair	t -1.56	low~high t +1.2
hi	zero	50	255.862	255.845	255.911	0.0143	+1.35
hi	one	50	255.861	255.846	255.914	0.0164	+1.36
hi	ones	50	255.862	255.848	255.911	0.0137	+1.27
hi	topbit	50	255.862	255.846	255.912	0.0148	+1.24
hi	alt	50	255.862	255.847	255.917	0.0156	+1.37
hi	lowhw	50	255.861	255.846	255.927	0.0156	+1.02
hi	highhw	50	255.863	255.846	255.929	0.0177	+1.22
hi	random_a	50	255.859	255.842	255.913	0.0129	-0.15
hi	random_b	50	255.858	255.842	255.902	0.0114	-1.06
hi	fixed_a	50	255.860	255.841	255.905	0.0115	+0.00
hi	fixed_b	50	255.861	255.847	255.929	0.0199	+1.16
hi	A/A random_a~random_b		t +0.85		fixed pair	t -1.16	low~high t -0.3
neg	zero	50	13.196	13.173	13.245	0.0105	-1.80
neg	one	50	13.198	13.184	13.230	0.0094	-0.60
neg	ones	50	13.200	13.183	13.222	0.0073	-0.76
neg	topbit	50	13.200	13.184	13.219	0.0065	-0.41
neg	alt	50	13.197	13.183	13.236	0.0107	-0.42
neg	lowhw	50	13.200	13.183	13.223	0.0085	-0.58
neg	highhw	50	13.201	13.186	13.238	0.0086	-0.68
neg	random_a	50	13.199	13.182	13.232	0.0096	-0.56
neg	random_b	50	13.202	13.184	13.225	0.0079	+0.83
neg	fixed_a	50	13.200	13.184	13.222	0.0085	+0.00
neg	fixed_b	50	13.200	13.184	13.222	0.0094	-0.45

mode	class	n	median c/op	min	max	sd	t vs fixed_a
neg	A/A random_a~random_b		t -1.36		fixed pair	t +0.45	low~high t +0.1
pos	zero	50	31.910	31.897	31.960	0.0164	+0.54
pos	one	50	105.442	104.906	105.531	0.1471	+3477.99
pos	ones	50	31.910	31.897	31.953	0.0100	-0.50
pos	topbit	50	105.409	104.915	105.503	0.1968	+2603.85
pos	alt	50	31.907	31.897	31.961	0.0122	-0.66
pos	lowhw	50	257.400	257.368	257.434	0.0146	+75000.51
pos	highhw	50	353.586	353.464	353.781	0.0736	+29966.83
pos	random_a	50	451.008	450.709	451.230	0.1026	+28295.28
pos	random_b	50	450.966	450.800	451.191	0.0766	+37548.82
pos	fixed_a	50	31.909	31.896	31.963	0.0151	+0.00
pos	fixed_b	50	31.910	31.896	31.969	0.0127	-0.60
pos	A/A random_a~random_b		t +1.49		fixed pair	t +0.60	low~high t -8973.3